

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

(192325096)

I V. Sandeep Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V. Sandeep Reddy
Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- Deduce the most likely original message.
- Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- Demonstrate how the encryption process transforms the message step-by-step.
- Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- Analyze how investigators could detect the presence of hidden data within the image.
- Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Secure Messaging System

a. Step-by-Step Encryption Process

This messaging system a polyalphabetic substitution cipher (then, Vigenere cipher). To encrypt the message, we assign each letter of the alphabet a numerical value from 0 to 25 ($A=0, B=1, C=2 \dots Z=25$).

We repeat the keyword to match the length of the message and then add their numerical value together. If the sum exceeds 25, we wrap around the alphabet.

Message: HELLO

Keyword: KEYKE

Plaintext	Keyword	Plaintext	Keyword	Sum(P+K)	Module	Ciphertext
H	K	7	10	17	17	R
E	E	4	4	8	8	I
L	Y	11	24	35	9	J
L	K	11	10	21	21	V
O	E	14	4	18	18	S

b. How the Repeating keyword influences the ciphertext

- Destroys letter patterns: Is a standard a different shift to each character "HELLO" would encrypt to double letter
- The power of the shift: Because the keyword repeats. The first 'L' is shifted by the letter 'Y' becoming a 'J'. The second 'L' is shifted by the letter 'K' becoming a 'V'.
- Security: This completely hides standard English letter frequencies, making it incredibly difficult for an attacker.

Suspicious Digital Asset Investigation

a. Detecting Hidden Data within an image

- **Statistical Analysis:-** Hiding data in an image's least significant bits predictably alters the natural distribution of pixel values. Investigators analyze the image's histogram to find unnatural clustering or anomalous pairings of odd and even pixel values that include tampering.
- **Visual Extraction:-** Investigator. Isolate and enhance the least significant bits of the image pixels in a natural image, this layer looks like random noise, if data is hidden sequentially, this layer will often reveal distinct structural blocks, patterns or unnatural artifacts.
- **Structural and Metadata:-** Examiners check the file format for anomalies, such as unexpected data appended past the stranded End-of-file (EOF) marker, missing colour palettes, or inconsistencies in the EXIF meta data.

b. Techniques to Strengthen Hidden communications

- To protect the embedded payload from detection and extraction, a sender can apply these two techniques.
- **Pre-Encryption:-** The payload should be strongly encrypted before being embedded into the logo. Even if investigators successfully detect the steganography and extract the payload bits, the resulting will be incomprehensible without the decryption key.

Randomized Scattering:- Instead of embedding the data sequentially starting from the top-left pixel. The sender can use data bits randomly number the image. This drastically reduces the statistical footprint in any one area of the image and makes extraction impossible without the exact key and algorithm used to determine the pixel locations.